



Pharynx & Esophagus — MCQ Practice

Anatomy GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A radiologist traces the pharynx on a lateral neck film. At which vertebral level does it end by becoming continuous with the esophagus?

EASY

- A. Lower border of C3
- B. Lower border of C4
- C. Upper border of C5
- D. Lower border of C6
- E. Upper border of T2

2. Which epithelium lines the nasopharynx, reflecting its purely respiratory function?

EASY

- A. Pseudostratified ciliated columnar
- B. Stratified squamous non-keratinized
- C. Simple columnar with goblet cells
- D. Stratified squamous keratinized
- E. Transitional epithelium

3. Which muscle forms the upper esophageal sphincter?

EASY

- A. Superior constrictor
- B. Thyropharyngeus
- C. Cricopharyngeus
- D. Palatopharyngeus
- E. Stylopharyngeus

4. What is the approximate length of the adult esophagus?

EASY

- A. 10 cm
- B. 15 cm
- C. 40 cm
- D. 6 m
- E. 25 cm (10 inches)

5. A 5-year-old boy has chronic mouth breathing, loud snoring and persistent nasal obstruction. Examination shows hypertrophy of the lymphoid tissue in the roof of the nasopharynx. Which tonsil is enlarged?

MEDIUM

- A. Palatine tonsil
- B. Pharyngeal tonsil
- C. Lingual tonsil
- D. Tubal tonsil
- E. Laryngeal tonsil

6. A 7-year-old child has a sore throat, high fever and painful swallowing. Oral examination shows enlarged palatine tonsils. Between which two folds do these tonsils lie?

MEDIUM

- A. Salpingopharyngeal and palatopharyngeal folds
- B. Palatoglossal and salpingopharyngeal folds
- C. Superior and middle constrictors
- D. Palatoglossal and palatopharyngeal arches
- E. Aryepiglottic and lateral glossoepiglottic folds

7. A fish bone lodged in the piriform fossa is removed carelessly. Afterwards the patient does not cough when fluid touches the supraglottic larynx. Which nerve was damaged?

MEDIUM

- A. Internal laryngeal nerve
- B. Recurrent laryngeal nerve
- C. External laryngeal nerve
- D. Glossopharyngeal nerve
- E. Hypoglossal nerve

8. A 70-year-old man has progressive dysphagia and regurgitates undigested food. A barium swallow shows a pouch projecting from the posterior wall of the lower pharynx. Where is the weak area through which it herniated?

MEDIUM

- A. Between superior and middle constrictors
- B. Between middle constrictor and stylopharyngeus
- C. Between thyropharyngeus and cricopharyngeus (Killian's dehiscence)
- D. At the esophageal hiatus
- E. Lateral to the piriform fossa

9. An endoscopist notes a smooth external indentation of the esophagus 27.5 cm from the incisor teeth. Which structure causes it?

MEDIUM

- A. Arch of the aorta
- B. Cricopharyngeus
- C. Right main bronchus
- D. Left main bronchus
- E. Diaphragmatic hiatus

10. A 52-year-old man with liver cirrhosis has dilated veins in the lower esophagus. Which vein drains this segment into the portal system?

MEDIUM

- A. Azygos vein
- B. Hemiazygos vein
- C. Inferior thyroid vein
- D. Superior vena cava
- E. Left gastric vein

11. A 60-year-old patient has carcinoma of the cervical esophagus. To which lymph node group does this segment primarily drain?

MEDIUM

- A. Axillary nodes
- B. Deep cervical nodes
- C. Tracheobronchial nodes
- D. Posterior mediastinal nodes
- E. Left gastric nodes

12. During surgery on the cervical esophagus, the surgeon works on its anterior surface. Besides the trachea, which structure lies anterior to it and must be protected?

MEDIUM

- A. Recurrent laryngeal nerve
- B. Carotid sheath
- C. Thoracic duct
- D. Thyroid lobes
- E. Azygos vein

13. A patient with a stroke has weakness of the superior pharyngeal constrictor. Which symptom would you expect during swallowing?

MEDIUM

- A. Aspiration into the larynx from the piriform fossa
- B. Air entering the esophagus during breathing
- C. Nasopharyngeal regurgitation
- D. Gastro-esophageal reflux
- E. Formation of a pharyngeal pouch

14. On a lateral neck X-ray, which vertebral bodies support the posterior wall of the oropharynx?

MEDIUM

- A. C1 and C2
- B. C3 to C6
- C. C6 and C7
- D. Body of C2 and upper part of C3
- E. C4 and C5 only

15. A patient has a large sliding hiatal hernia that carries the gastro-esophageal junction up into the thorax. Which two external components of the anti-reflux barrier are lost?

HARD

- A. The cricopharyngeus and the superior constrictor
- B. The pyloric sphincter and the angle of His
- C. The left crus and the lesser omentum
- D. The upper esophageal sphincter and the right crus
- E. The right crus of the diaphragm (crural sphincter) and the angle of His

16. Patient A has heartburn because gastric contents reflux into the esophagus. Patient B has dysphagia with a dilated esophagus above a tight gastro-esophageal junction. Which statement correctly contrasts the two?

HARD

- A. A: reduced LES pressure or inappropriate transient relaxation; B: failure of the LES to relax (achalasia)
- B. A: failure of the LES to relax; B: reduced LES pressure
- C. Both are caused by weakness of the cricopharyngeus
- D. A: Killian's dehiscence; B: hiatal hernia
- E. Both result from compression by the left main bronchus

17. A tumour arising from the RIGHT wall of the mid-thoracic esophagus spreads directly outward. Which structure would it meet first?

HARD

- A. Aortic arch
- B. Mediastinal pleura
- C. Left subclavian artery
- D. Descending thoracic aorta
- E. Left recurrent laryngeal nerve

18. During a total gastrectomy the left gastric artery is ligated at its origin. Which part of the esophagus loses part of its blood supply, and which other artery still supplies it?

HARD

- A. Cervical part; inferior thyroid artery
- B. Thoracic part; bronchial arteries
- C. Abdominal part; left inferior phrenic artery
- D. Thoracic part; esophageal branches of the descending aorta
- E. Abdominal part; right gastric artery

19. Why do the submucosal veins of the lower esophagus become varices in portal hypertension, while those of the mid-thoracic esophagus usually do not?

HARD

- A.** The mid-thoracic esophagus has no submucosal veins
- B.** The mid-thoracic veins drain into the inferior thyroid vein
- C.** The lower esophagus is supplied by the superior mesenteric vein
- D.** Mid-thoracic veins are protected by the right crus
- E.** Lower esophageal veins drain both to the left gastric (portal) and azygos (systemic) veins, forming a porto-systemic anastomosis

20. An endoscope is advanced from the incisors. The endoscopist records narrowings at 15 cm, 22.5 cm, 27.5 cm and 40 cm. Which option correctly matches each distance to its cause?

HARD

- A.** Cricopharyngeus; left main bronchus; aortic arch; diaphragm
- B.** Cricopharyngeus; aortic arch; left main bronchus; diaphragm
- C.** Superior constrictor; aortic arch; right main bronchus; LES
- D.** Cricopharyngeus; aortic arch; left main bronchus; pyloric sphincter
- E.** Thyropharyngeus; left main bronchus; aortic arch; diaphragm

✓ Answer key

Q	Answer	Difficulty	Why
1	D	EASY	The pharynx runs from the base of the skull to the lower border of C6, where it continues as the esophagus at the level of the cricoid cartilage.
2	A	EASY	The nasopharynx resembles the nose. It is respiratory, lined by pseudostratified ciliated columnar epithelium, and has rigid, non-collapsible walls.
3	C	EASY	The cricopharyngeus part of the inferior constrictor is the anatomical, striated upper esophageal sphincter. It keeps air out of the esophagus during breathing.
4	E	EASY	The esophagus is about 25 cm (10 in) long: 4 cm cervical, 20 cm thoracic and 1.5–3 cm abdominal. 40 cm is the distance from the incisors to the GE junction.
5	B	MEDIUM	The pharyngeal (nasopharyngeal) tonsil, or adenoid, lies postero-superiorly in the nasopharyngeal roof, the part of Waldeyer's ring that obstructs the nasal airway when enlarged.
6	D	MEDIUM	The palatine tonsil sits in the tonsillar fossa on the lateral wall of the oropharynx, with the palatoglossal arch in front and the palatopharyngeal arch behind.
7	A	MEDIUM	The internal laryngeal nerve pierces the thyrohyoid membrane and runs under the mucosa of the floor of the piriform fossa. Injury causes supraglottic anaesthesia and loss of the cough reflex.
8	C	MEDIUM	Killian's dehiscence is a triangular weak area in the posterior pharyngeal wall between the two parts of the inferior constrictor. A pharyngeal pouch forms here, with dysphagia.
9	D	MEDIUM	The thoracic (broncho-aortic) constriction is compound: the aortic arch at 22.5 cm (T4), then the left main bronchus at 27.5 cm (T5).
10	E	MEDIUM	The abdominal esophagus drains to the portal circulation via the left gastric vein and to the systemic circulation via the azygos vein. This is the porto-systemic anastomosis.
11	B	MEDIUM	The cervical esophagus drains into the deep cervical nodes. The thoracic part drains to superior and posterior mediastinal nodes; the abdominal part to posterior mediastinal and left gastric nodes.
12	A	MEDIUM	Anterior relations of the cervical esophagus are the trachea and the recurrent laryngeal nerve. The thyroid lobes and carotid sheath are lateral; C6–C7 are posterior.
13	C	MEDIUM	The superior constrictor closes off the upper pharynx during swallowing. The lecture notes that its weakness may cause nasopharyngeal regurgitation.
14	D	MEDIUM	The posterior wall of the oropharynx lies on the body of C2 and the upper C3. The laryngopharynx lies on C3–C6.
15	E	HARD	The LES is physiological. Its action depends on the crural diaphragm (the right crus around the hiatus) and the acute angle of His, which acts as a flap valve. Both are lost when the junction moves into the chest.
16	A	HARD	GERD results from reduced LES pressure or inappropriate transient LES relaxations. Achalasia of the cardia is failure of the gastro-esophageal sphincter to relax, obstructing flow into the stomach.
17	B	HARD	The right side of the thoracic esophagus is related only to the mediastinal pleura. The left side has the left subclavian artery, aortic arch and thoracic duct; posteriorly lie the vertebrae, thoracic duct, azygos veins and descending aorta.
18	C	HARD	The abdominal esophagus is supplied by the left gastric artery (celiac trunk) and the left inferior phrenic artery. The cervical part relies on the inferior thyroid artery; the thoracic part on aortic esophageal branches.
19	E	HARD	Only the abdominal segment drains to both systems. Raised portal pressure is transmitted through the left gastric vein into this anastomosis. The thoracic veins drain only into the azygos system.
20	B	HARD	15 cm (C6) cricopharyngeus; 22.5 cm (T4) aortic arch; 27.5 cm (T5) left main bronchus; 40 cm diaphragmatic hiatus. Mnemonic ABCD: Aortic arch, Bronchus, Cricopharyngeus, Diaphragm.